

Containers docker

do desenvolvimento para produção

Vitor Mattos

Quem sou eu?

Realizador de sonhos desde 2003

Amante de opensource

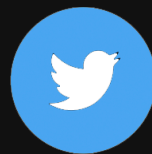
Palestrante

PHP Zend Certified Engineer ([ZEND024235](#))

PHPRio (<https://t.me/phprio>)

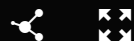
CTO LibreCode coop (<https://t.me/LibreCodeCoop>)

Redes sociais: (VitorMattos ou VitorMattosRJ)



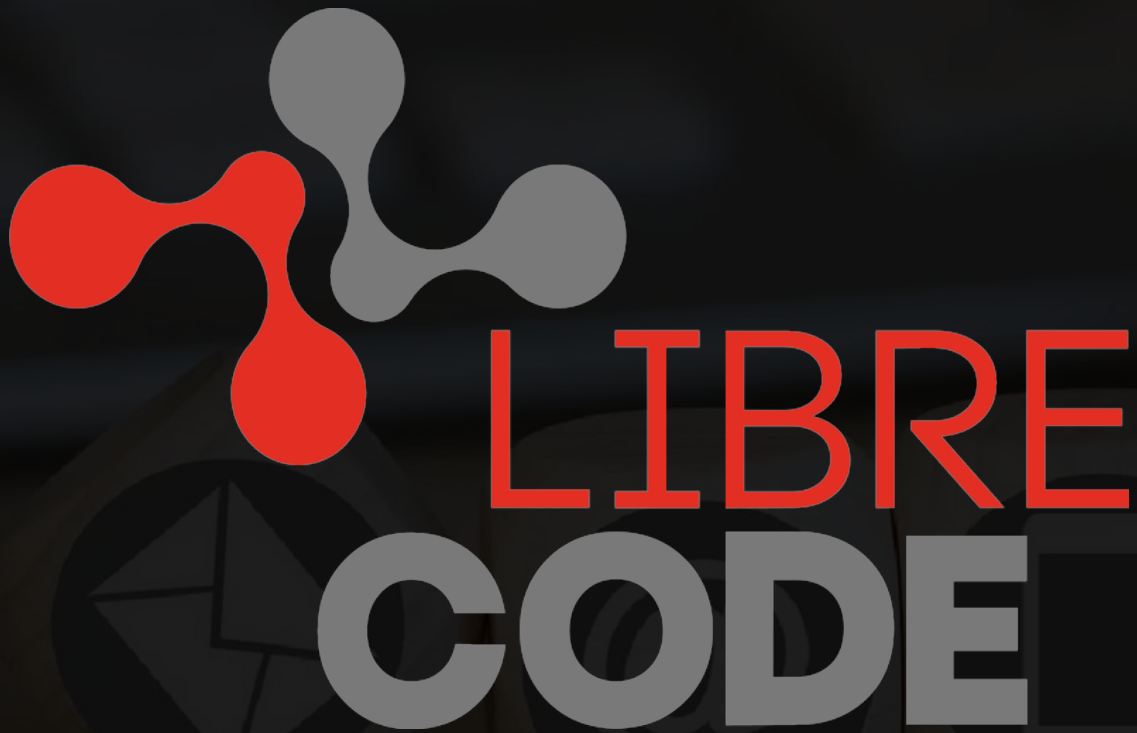


By Vitor Mattos



Porque Docker?





coperativa de TI



Agilidade para montar ambiente





Ambiente replicável

A low-angle, upward-looking shot of numerous yellow umbrellas. The umbrellas are packed closely together, creating a dense, textured surface of yellow and orange segments. The tips of the black umbrella handles are visible, pointing upwards. The background is a bright blue sky filled with soft, white, wispy clouds. A semi-transparent black banner with rounded ends is centered horizontally across the middle of the image, containing white text.

Redução de diferenças entre setup

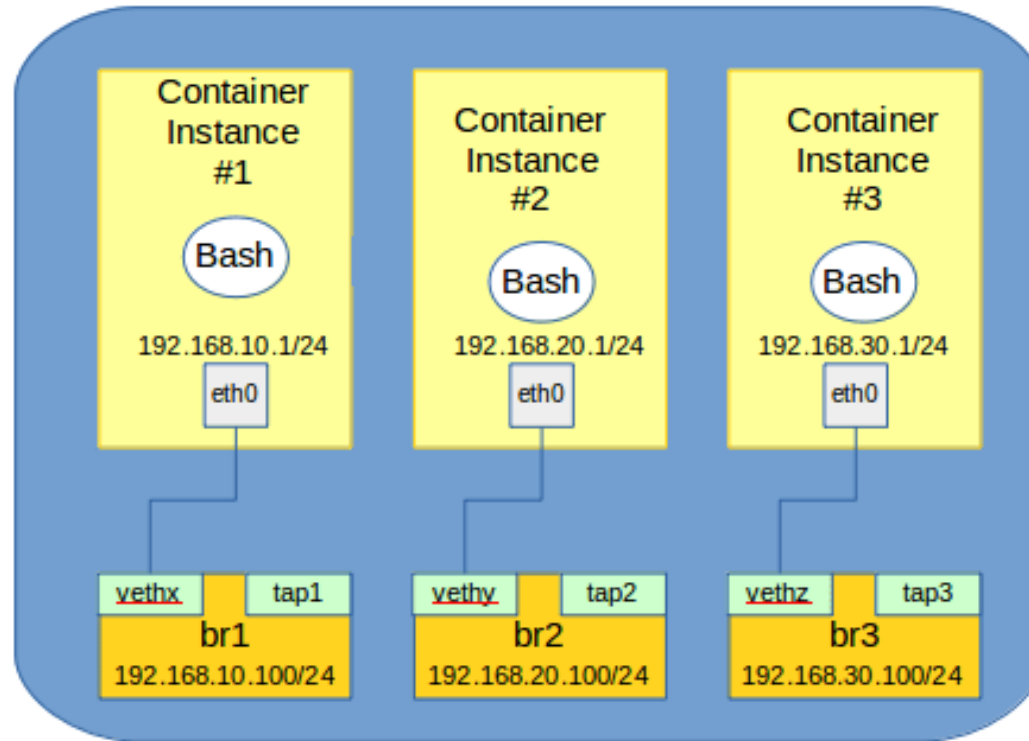
Uso de diferentes versões

Possibilidade de uso de diferentes versões
de tudo no mesmo computador/servidor



Isolamento de processos

Single host



Ambiente de desenvolvimento

```
15 | Register Controller
16 |
17 |
18 | This controller handles the registration of new users, including
19 | validation and creation. By default, this controller is not
20 | provide this functionality without requiring the user to be
21 |
22 | */
23 |
24 | use RegistersUsers;
25 |
26 | /**
27 |
28 | * @var string
29 | */
30 | protected $redirectTo = '/home';
31 |
32 | /**
33 |
34 | * Create a new controller instance.
35 | *
36 | * @return void
37 | */
```

3 javascript | app.js

Porque Docker em desenvolvimento?

Simplicidade

Porque Docker em desenvolvimento?

Documentação

Produção

Talk is cheap
Show me the code!

<https://github.com/librecodecoop>



CI: Teste suas imagens

ShellCheck

finds bugs in your shell scripts.

CI: Teste suas imagens

GitHub action for ShellCheck

CI: Teste suas imagens

Dependabot



CI: Teste suas imagens

Dependabot

```
1 version: 2
2 updates:
3
4   - package-ecosystem: "github-actions"
5     directory: "/"
6     schedule:
7       interval: "daily"
8   - package-ecosystem: "docker"
9     directory: "/.docker/"
10    schedule:
11      interval: "daily"
```



Registry

Repositório de imagens

Why a registry?!

Quick deploy



CI: Disponibilize para o mundo

Build and push Docker images

```
1 - name: Login to Docker Hub
2   uses: docker/login-action@v3
3   with:
4     username: ${ secrets.DOCKERHUB_USERNAME }
5     password: ${ secrets.DOCKERHUB_TOKEN }
6 - name: Build and push
7   uses: docker/build-push-action@v5
8   with:
9     push: true
10    tags: user/app:latest
```

Exemplo real

CI: Disponibilize para o mundo

Build and push Docker images

```
1 services:
2   nextcloud:
3     image: ghcr.io/librecodecoop/nextcloud-dev-php${PHP_VERSION:-82}:latest
```

Exemplo real

Vamos para o servidor?!

Estrutura de pastas

Exemplo

```
`-- projects
    |-- akaunting
    |-- archive
    |-- duplicati
    |-- mysql
    |-- nextcloud-docker
    |-- nginx-proxy
    |-- onlyoffice
    |-- postgres
    |-- producao-cooperativista
    |-- redis
    `-- site
```

Nginx proxy

Let's Encrypt

Show me the code

Database

```
1 networks:
2   mysql:
3     external: true
4     name: mysql
5   reverse-proxy:
6     external: true
7     name: reverse-proxy
8
9 services:
10  mysql:
11    build: ../docker/mysql
12    volumes:
13      #- ../docker/mysql/conf:/etc/mysql/conf.d
14      - ../docker/volumes/mysql/dump:/docker-entrypoint-initdb.d
15      - ../docker/volumes/mysql/data:/var/lib/mysql
16    environment:
17      - MYSQL_ROOT_PASSWORD
18    networks:
19      - mysql
20    restart: always
```

Backup

```
1 networks:
2   reverse-proxy:
3     external: true
4     name: reverse-proxy
5
6 services:
7   duplicati:
8     image: duplicati/duplicati
9     container_name: duplicati
10    restart: always
11    volumes:
12      - ./volumes/duplicati-data:/data
13      - /root/projects:/projects
14    environment:
15      - VIRTUAL_HOST
16      - LETSENCRYPT_HOST
17      - LETSENCRYPT_EMAIL
18      - VIRTUAL_PORT=8200
```

Monitoramento



Zabbix docker

Conclusão

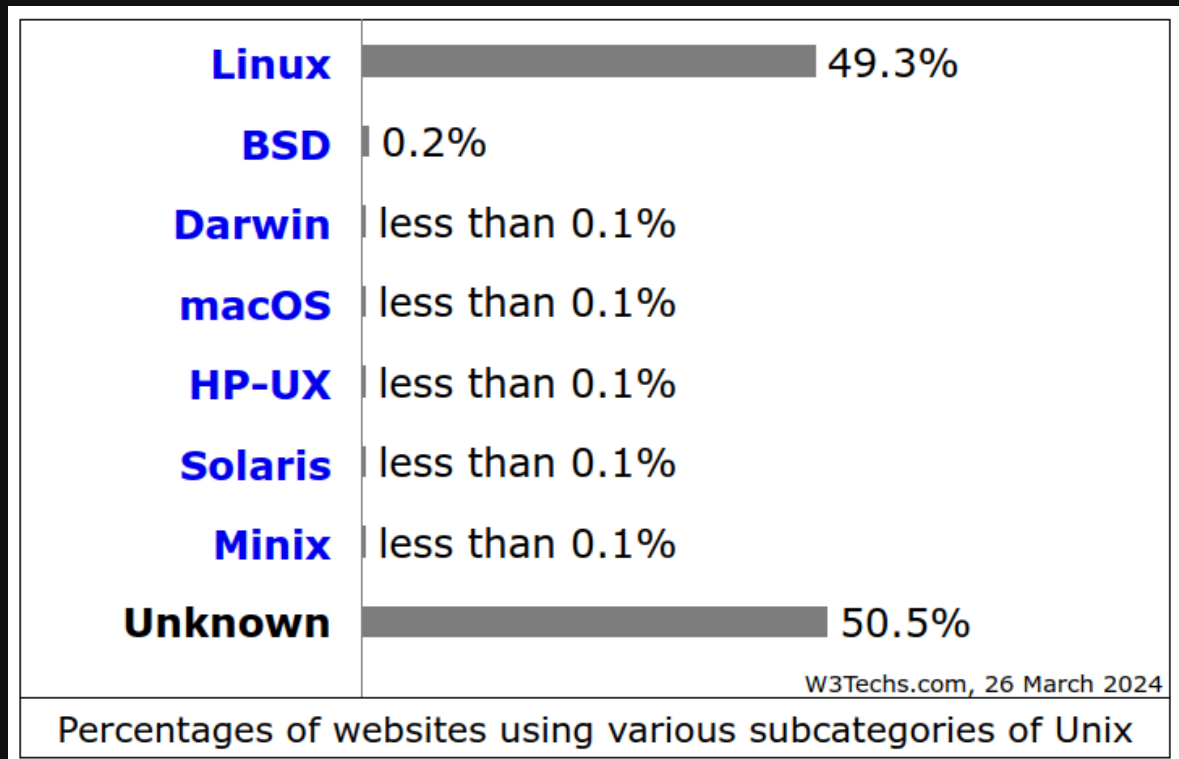
Conclusão

Use Linux



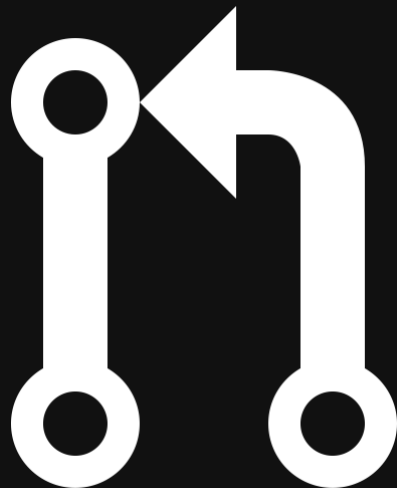
Conclusão

Use Linux

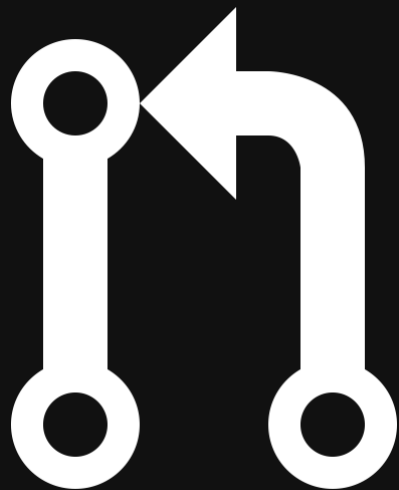


E no código, como faço?

Compartilhe seu conhecimento



Contribua com projetos de código aberto

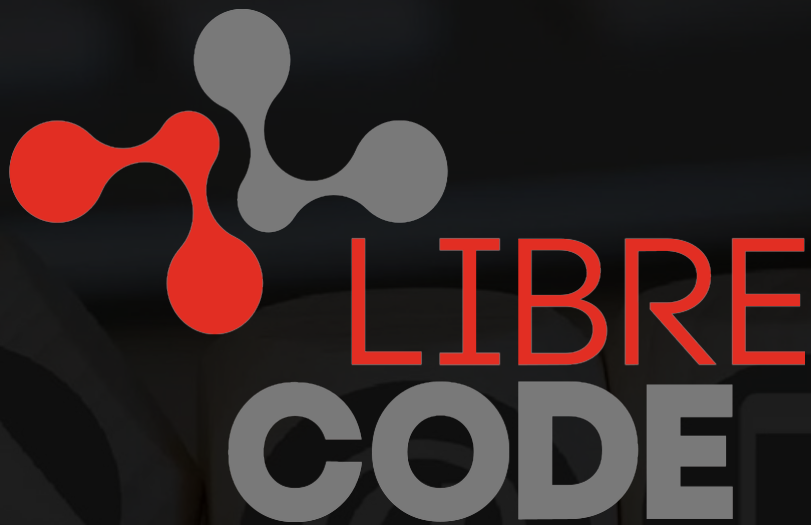


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Conheça mais

- Plataforma CapacitaCoop
- SESCOOP





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